TEDTRONICS · PERFBOARD FALLBACK

CP00 — First light

Win: the ESP32 switches the real four-column LED loop.

85	44	1.7 m
SOLDER POINTS	WIRES TO SOLDER	WIRE TO CUT

9	24	5
JUMPERS	AWG WIRE	GROUPS

WARNING

Disconnect USB and all power sources before you build. Do not connect the final supply or the matrix in CP00.

BUILD ORDER

1. **Pogo contact columns** — 4 parts · 20 wires · 24 solder points — p.3
2. **NODE header and wires** — 1 parts · 4 wires · 14 solder points — p.4
3. **ESP32 controller** — 3 parts · 11 wires · 47 solder points — p.5
4. **Breadboard USB switch** — 8 parts · 9 jumpers — p.6
5. **Fit the kit and test** — 4 kit parts · 6 checks · film — p.7

TOOLS

- [ ] Soldering iron and solder
- [ ] Wire stripper
- [ ] USB cable
- [ ] Wire cutter
- [ ] Multimeter
- [ ] Sleeves for wire crossings

WIRE

- Insulated 24 AWG, all hook-up wires. Run them under the board.
- Bare wire for the 20 node straps*.
- Use the wire colors on the cut lists.

* The guidelines give a size only for insulated wire. Use 24 AWG bare wire and confirm.

READ THIS ONCE

Key

Symbols used on every page.



Red pin
Pin 1 of a part. Match it to pin 1 on the part.



Open pin
Any other pin. Top view only.



Solder pad
Bottom view. Make a solder joint here.



Insulated wire
Under the board. Solder only the two ends. Put a sleeve on every crossing.



Bare strap
Joins the pins of one column. Touch no other column.



Part body
Top view. Colored = this group. Dashed gray = built earlier.



Row tie
Breadboard. The row already joins these holes. Add no jumper.

TWO VIEWS OF A PERFBOARD

Numbers on the edges are the **top** hole numbers. They are the same on both views.

- **TOP**: look at the part side.
- **BOTTOM (mirrored)**: turn the board over left-to-right. Column 1 is now on the right.

WORDS

WARNING

A warning tells you about a risk to people or to the board. Read it first.

CAUTION

A caution tells you about a risk to a part or to a measurement.

NOTE

A note gives extra information.

GROUP 1 OF 5

Pogo contact columns

Board A1 · four columns of six contacts

4	24	20
POGO PARTS	SOLDER POINTS	BARE STRAPS

NOTE

All four columns stay electrically separate. Each column is one node.

PARTS

REF	PART	PINS
J_NODE00	NODE00 — six contacts	**6**
J_NODE01	NODE01 — six contacts	**6**
J_NODE02	NODE02 — six contacts	**6**
J_NODE03	NODE03 — six contacts	**6**

CUT LIST

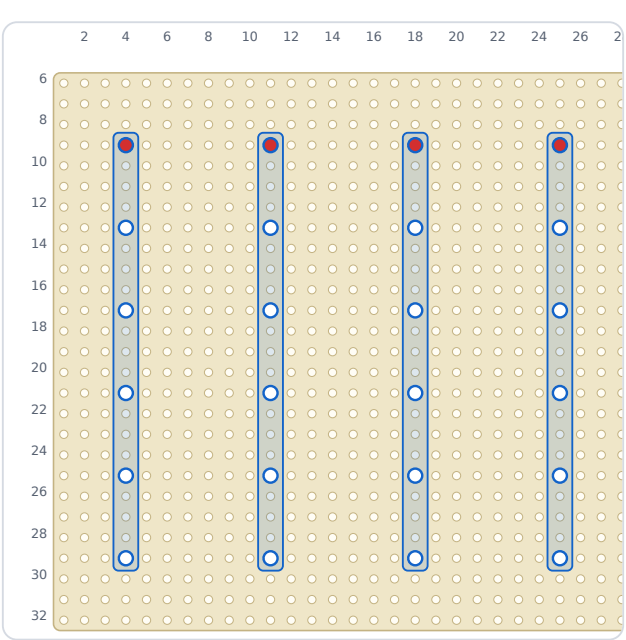
20 × 17 mm

Bare wire, 24 AWG. 4 pad gaps (10.2 mm) plus 6 mm for the ends.

VIEWS

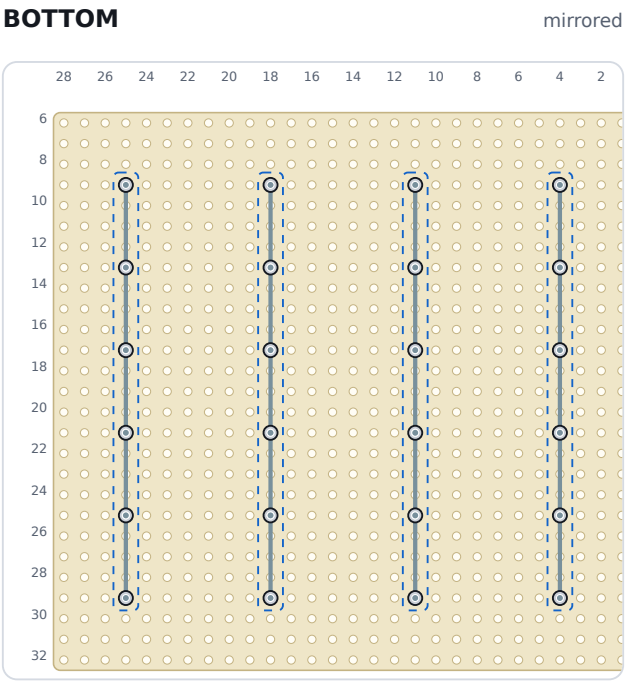
TOP

pogo side



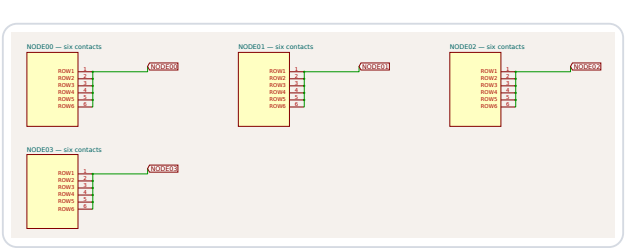
BOTTOM

mirrored



Schematic

tscircuit



STEPS

1 **Install the pogo pins.** Put each part in from the top, pogo side up.

- [ ] J_NODE00: column 4, rows 9, 13, 17, 21, 25, 29
- [ ] J_NODE01: column 11, rows 9, 13, 17, 21, 25, 29
- [ ] J_NODE02: column 18, rows 9, 13, 17, 21, 25, 29
- [ ] J_NODE03: column 25, rows 9, 13, 17, 21, 25, 29

2 **Solder the pins.** Turn the board over. Solder all 24 pins from the bottom.

- [ ] Each pin is straight and square to the board.
- [ ] No solder bridge to the next column.

3 **Cut 20 bare straps**, each 17 mm.

4 **Solder the straps.** In each column, join pin 1–2, 2–3, 3–4, 4–5, 5–6 on the bottom.

- [ ] Column 4: 5 straps
- [ ] Column 11: 5 straps
- [ ] Column 18: 5 straps
- [ ] Column 25: 5 straps

5 **Measure with the multimeter in continuity mode.** Power OFF.

- [ ] All 6 pins in one column read below 1 Ω.
- [ ] Pins in two different columns read open (above 10 MΩ).

GROUP 2 OF 5

NODE header and wires

Board A1 · header and four blue wires

10	4	14
HEADER PINS	WIRES	SOLDER POINTS

CAUTION

Mount the header on the **bottom** side. Pin 1 is at column 3, row 33.

PARTS

REF	PART	PINS
J_NODE	NODE cable → matrix	**10**

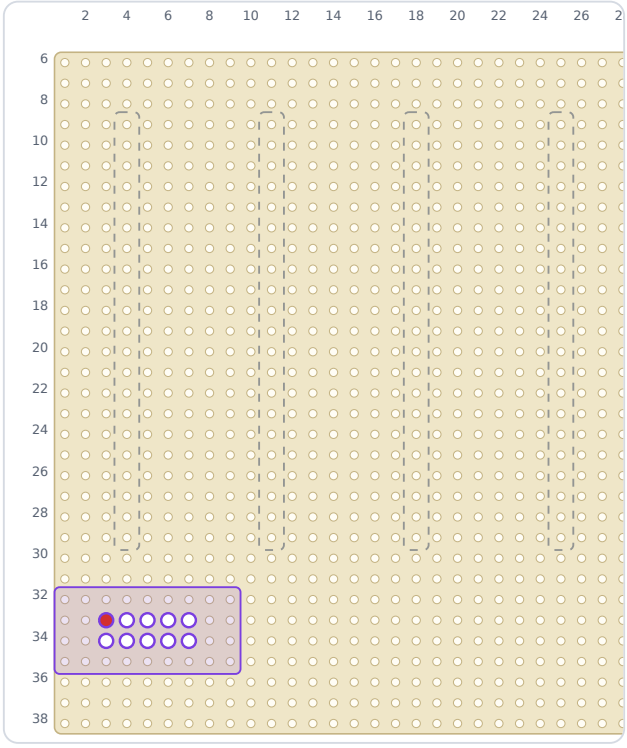
CUT LIST

- **1** × 79 mm
- **1** × 94 mm
- **1** × 110 mm
- **1** × 125 mm

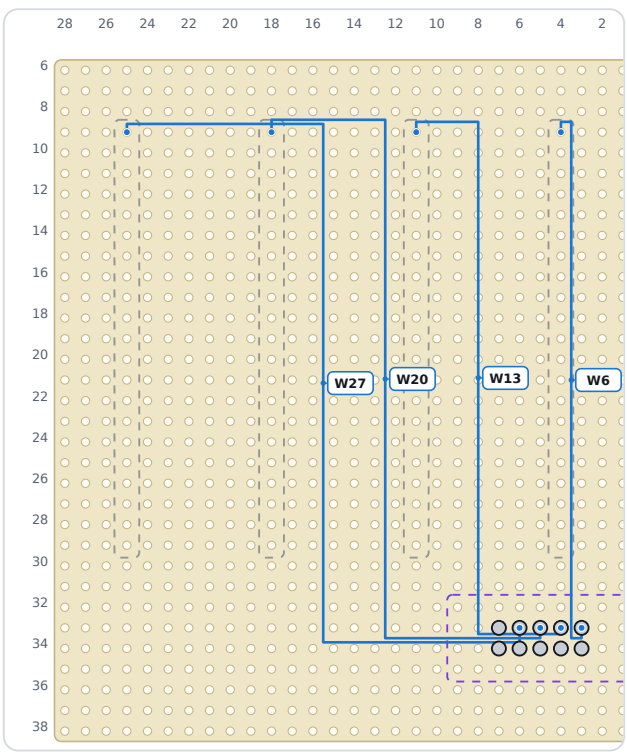
Insulated 24 AWG. Routed length plus 10 mm for the ends.

VIEWS

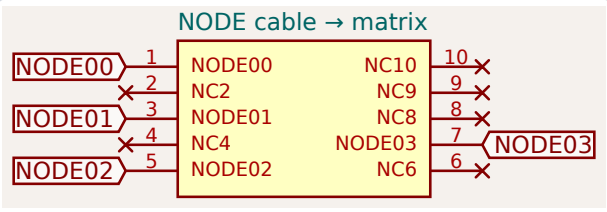
TOP context



BOTTOM mirrored



Schematic tscircuit



WIRES

ID	FROM → TO	CUT
W6	J_NODE00.1 → J_NODE.1 holes (4,9) → (3,33)	**79 mm**
W13	J_NODE01.1 → J_NODE.3 holes (11,9) → (4,33)	**94 mm**
W20	J_NODE02.1 → J_NODE.5 holes (18,9) → (5,33)	**110 mm**
W27	J_NODE03.1 → J_NODE.7 holes (25,9) → (6,33)	**125 mm**

STEPS

1. **Install the J_NODE header** on the bottom side. Pin 1 at column 3, row 33.
   - [ ] Red pad is at pin 1.
2. **Solder the 10 header pins** from the top side.
   - [ ] No solder bridge between pins.
3. **Cut 4 wires:** **1** × 79 mm · **1** × 94 mm · **1** × 110 mm · **1** × 125 mm
4. **Route and solder the wires on the bottom.** Solder only the two ends of each wire.
   - [ ] W6: J_NODE00.1 → J_NODE.1, 79 mm
   - [ ] W13: J_NODE01.1 → J_NODE.3, 94 mm
   - [ ] W20: J_NODE02.1 → J_NODE.5, 110 mm
   - [ ] W27: J_NODE03.1 → J_NODE.7, 125 mm
5. **Put a sleeve on every crossing.** A crossing is not a connection.
6. **Measure.** Power OFF.
   - [ ] Each wire reads below 1 Ω from end to end.
   - [ ] Two different wires read open.

GROUP 3 OF 5

ESP32 controller

Matrix board · ESP32 and two headers

47	11	0.94 m
SOLDER POINTS	WIRES	WIRE

WARNING

Disconnect USB before you solder or measure.

CAUTION

Check your ESP32 DevKit pinout before you solder. Pin labels must match the table. Measure the real board first.

PARTS

REF	PART	PINS
U_ESP1	ESP32 DevKitC-38 — router ESP32-DevKitC V4 ESP32-WROOM-32E	**38**
J_FIXTURE	manual continuity fixture; ESP1 GPIO4 SDA / GPIO16 SCL / GPIO25 stimulus	**5**
J_PROTOTYPE	USB breadboard prototype; remove before final rail connection	**4**

CUT LIST

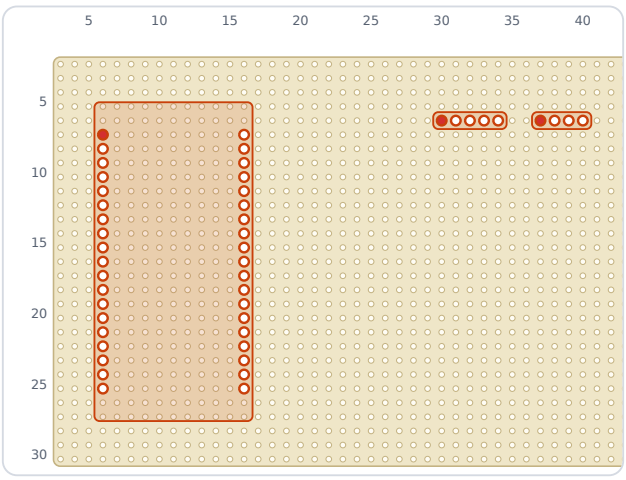
- **1** × 36 mm
- **1** × 36 mm
- **1** × 56 mm
- **1** × 71 mm
- **1** × 79 mm
- **2** × 89 mm
- **1** × 110 mm
- **1** × 115 mm
- **1** × 120 mm
- **1** × 143 mm

Insulated 24 AWG. Routed length plus 10 mm for the ends.

VIEWS

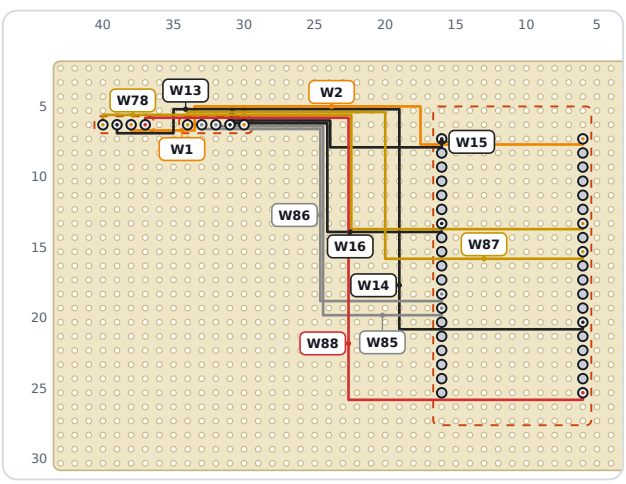
TOP

part side



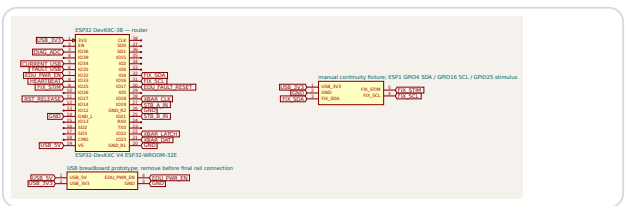
BOTTOM

mirrored



Schematic

tscircuit



WIRES

ID	FROM → TO	CUT
W1	J_FIXTURE.1 → J_PROTOTYPE.2 holes (30,6) → (38,6)	**36 mm**
W2	J_FIXTURE.1 → U_ESP1.1 holes (30,6) → (6,7)	**79 mm**
W13	J_FIXTURE.2 → J_PROTOTYPE.3 holes (31,6) → (39,6)	**36 mm**
W14	J_FIXTURE.2 → U_ESP1.14 holes (31,6) → (6,20)	**115 mm**
W15	J_FIXTURE.2 → U_ESP1.20 holes (31,6) → (16,7)	**56 mm**
W16	J_FIXTURE.2 → U_ESP1.26 holes (31,6) → (16,13)	**71 mm**
W78	J_PROTOTYPE.4 → U_ESP1.7 holes (40,6) → (6,13)	**120 mm**
W85	J_FIXTURE.3 → U_ESP1.32 holes (32,6) → (16,19)	**89 mm**
W86	J_FIXTURE.4 → U_ESP1.31 holes (33,6) → (16,18)	**89 mm**
W87	J_FIXTURE.5 → U_ESP1.9 holes (34,6) → (6,15)	**110 mm**
W88	J_PROTOTYPE.1 → U_ESP1.19 holes (37,6) → (6,25)	**143 mm**

STEPS

1 **Install the ESP32 DevKit.**

- [ ] U_ESP1: column 6, rows 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25; column 16, rows 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25

2 **Install the two headers** on the top side. Red pad is at pin 1.

- [ ] J_FIXTURE: column 30, rows 6; column 31, rows 6; column 32, rows 6; column 33, rows 6; column 34, rows 6
- [ ] J_PROTOTYPE: column 37, rows 6; column 38, rows 6; column 39, rows 6; column 40, rows 6

3 **Solder all 47 pins** from the bottom.

- [ ] No solder bridge between pins.

4 **Cut 11 wires** from the cut list.

5 **Route and solder the wires on the bottom.** Solder only the two ends. Put a sleeve on every crossing.

- [ ] W1 J_FIXTURE.1 → J_PROTOTYPE.2
- [ ] W2 J_FIXTURE.1 → U_ESP1.1
- [ ] W13 J_FIXTURE.2 → J_PROTOTYPE.3
- [ ] W14 J_FIXTURE.2 → U_ESP1.14
- [ ] W15 J_FIXTURE.2 → U_ESP1.20
- [ ] W16 J_FIXTURE.2 → U_ESP1.26
- [ ] W78 J_PROTOTYPE.4 → U_ESP1.7
- [ ] W85 J_FIXTURE.3 → U_ESP1.32
- [ ] W86 J_FIXTURE.4 → U_ESP1.31
- [ ] W87 J_FIXTURE.5 → U_ESP1.9
- [ ] W88 J_PROTOTYPE.1 → U_ESP1.19

6 **Measure.** Power OFF.

- [ ] Each wire reads below 1 Ω from end to end.
- [ ] GND and 3V3 read open.

GROUP 4 OF 5

Breadboard USB switch

Breadboard · default-OFF USB switch

8	9	0
PARTS	JUMPERS	SOLDER POINTS

WARNING

Keep the USB cable disconnected. Put in all parts first.

CAUTION

Q_PASS and Q_ENABLE have three pins in a fixed order. Read the pin letters on the drawing.

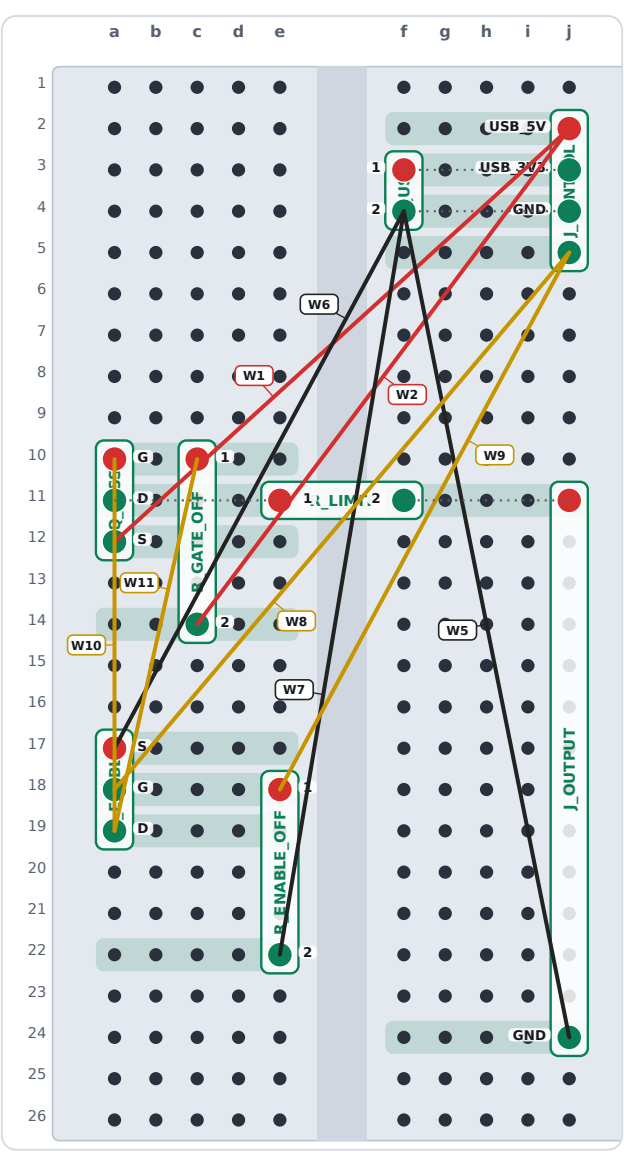
PARTS

REF	PART	PINS
J_CONTROL	Four individual flying leads from ESP1 prototype header	4
Q_PASS	USB high-side switch NDP6020P	3
Q_ENABLE	3.3V gate-control transistor 2N7000	3
R_GATE_OFF	10kΩ makes the P-channel switch OFF · 10k MFR-25FBF52-10K	2
R_ENABLE_OFF	100kΩ holds enable LOW · 100k MFR-25FBF52-100K	2
R_LIMIT	560Ω ±1%, 0.25W; bounds every educational-source path · 560 MFR-25FBF52-560R	2
J_OUTPUT	Two flying leads: C2 positive / C1 return	2
C_USB	USB3V3 fixture decoupling · 100nF C315C104K5R5TA	2

TOP VIEW

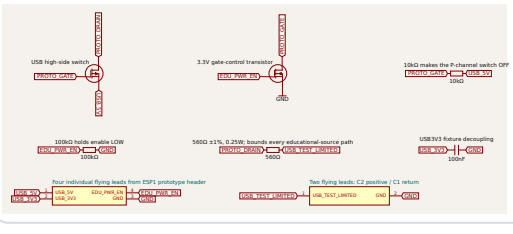
BREADBOARD

rows a–e and f–j are joined



Schematic

tscircuit



JUMPERS

ID		FROM → TO (HOLES)
W1	🔴	j2 → a12 USB_5V
W2	🔴	j2 → c14 USB_5V
W5	⚫	f4 → j24 GND
W6	⚫	f4 → a17 GND
W7	⚫	f4 → e22 GND
W8	🟡	j5 → a18 EDU_PWR_EN
W9	🟡	j5 → e18 EDU_PWR_EN
W10	🟡	a19 → a10 PROTO_GATE
W11	🟡	a19 → c10 PROTO_GATE

No jumper for 4 row ties. The breadboard rows already join them.

STEPS

1 **Put the parts in the holes.** Pin letters on the drawing must match.

- [ ] J_CONTROL: j2, j3, j4, j5
- [ ] Q_PASS: a10, a11, a12
- [ ] Q_ENABLE: a17, a18, a19
- [ ] R_GATE_OFF: c10, c14
- [ ] R_ENABLE_OFF: e18, e22
- [ ] R_LIMIT: e11, f11
- [ ] J_OUTPUT: j11, j24
- [ ] C_USB: f3, f4

2 **Add the 9 jumpers.** Use male jumper wire. No soldering.

- [ ] W1 🔴 j2 → a12
- [ ] W2 🔴 j2 → c14
- [ ] W5 ⚫ f4 → j24
- [ ] W6 ⚫ f4 → a17
- [ ] W7 ⚫ f4 → e22
- [ ] W8 🟡 j5 → a18
- [ ] W9 🟡 j5 → e18
- [ ] W10 🟡 a19 → a10
- [ ] W11 🟡 a19 → c10

3 **Check the 4 row ties.** USB disconnected.

- [ ] Each tie reads below 1 Ω.

4 **Check pin 1 and polarity** of every part.

GROUP 5 OF 5

Fit the kit and test

No soldering · record every reading

4	6	1
KIT PARTS	CHECKS	FILM

WARNING

Do not use a GPIO pin as a battery. The kit source is the 560 Ω limited USB test source.

NOTE

The firmware is not in this guide. You need the compatible factory firmware revision for check 5.

KIT PARTS ON BOARD A1

ROW	PART	ON
1 · y 9	Fake battery	− C1, + C2
2 · y 13	Resistor 1 kΩ	C2 – C3
3 · y 17	LED	anode C3, cathode C4
4 · y 21	Kit jumper	C4 → C1

C1–C4 are columns 1–4 (NODE00–NODE03).

STEPS

1 **Link the test source to A1.** Use two flying leads from the breadboard J_OUTPUT.

- ☐ J_OUTPUT pin 1 → A1 J_NODE pin 3 (C2, positive).
- ☐ J_OUTPUT pin 2 → A1 J_NODE pin 1 (C1, ground).

2 **Connect the USB prototype leads.** Four separate leads from matrix J_PROTOTYPE to breadboard J_CONTROL.

- ☐ Pin 1 matches pin 1 on both ends.

3 **Fit the four kit parts** on the contact rows. See the table above.

- ☐ LED polarity is correct.
- ☐ Measured resistor value is 1 kΩ.

CHECKS

1 **Identify the hardware.** Find the ESP32 DevKit pinout, USB 5V and 3V3. Measure the real pogo and adapter size before final solder.

- ☐ Pinout matches the table.

2 **Measure A1, unpowered.** Pins in one column: below 1 Ω. Different columns: above 10 MΩ.

Column result _____ pass / fail

3 **Measure the prototype, A1 disconnected.** USB 5V: 5.25 V or less. 3V3: 3.20 to 3.38 V. Short-circuit current: 9.5 mA or less.

USB 5V _____ V

3V3 _____ V

Short current _____ mA

CAUTION

Short the output only for the current test. Remove the short before you fit the kit.

4 **Run the firmware.** GPIO32 LOW: LED is OFF. Enable: LED is ON. Disable: LED is OFF.

Firmware revision _____

Voltage at C2, under load _____ V

NOTE

The test source is not regulated. Do not call it the final 5V rail.

5 **Do the negative test.** Reset the ESP32. The LED must go dark. Do not touch the kit.

- ☐ LED is dark after reset.

6 **Film the win.** Show power OFF and the firmware revision. Show the LED switch ON and OFF. Show the real readings.

- ☐ Record readings, units, equipment, link positions and firmware revision.

☐ **Accept CP00 — all checks pass**